

Research Notes on Multi-armed bandit

Multi-armed bandit

>> Section 1

Combinatorial bandit The Combinatorial Multiarmed Bandit (CMAB) problem arises when instead of a single discrete variable to choose from, an agent needs to choose values for a set of variables. Assuming each variable is discrete, the number of possible choices per iteration is exponential in the number of variables. Several CMAB settings have been studied in the literature, from settings where the variables are binary to more general setting where each variable can take an arbitrary set of values.

>> Section 2

Adversarial bandit Another variant of the multi-armed bandit problem is called the adversarial bandit, first introduced by Auer and Cesa-Bianchi (1998). In this variant, at each iteration, an agent chooses an arm and an adversary simultaneously chooses the payoff structure for each arm. This is one of the strongest generalizations of the bandit problem as it removes all assumptions of the distribution and a solution to the adversarial bandit problem is a generalized solution to the more specific bandit problems.

>> Section 3

Pricing strategies Pricing strategies establish a price for each lever. For example, as illustrated with the POKER algorithm, the price can be the sum of the expected reward plus an estimation of extra future rewards that will gain through the additional knowledge. The lever of highest price is always pulled.

>> Section 4

For each $t = 1, 2, \dots, T$. Set $p_i(t) = (1 - \gamma) \omega_i(t) \sum_{j=1}^K \omega_j(t) + \gamma K$
$$p_i(t) = (1 - \gamma) \frac{\omega_i(t)}{\sum_{j=1}^K \omega_j(t)} + \gamma \frac{K}{\sum_{j=1}^K \omega_j(t)}$$

$$i = 1, \dots, K$$

>> Section 5

Hence, the regret $\rho_\pi(T)$
$$\rho_\pi(T)$$
 for policy π π is computed as the difference between $D(T)$ ${\mathcal{D}}(T)$ and the cumulative expected reward at step T T for policy π π :

>> Section 6

$\rho = T \mu^* - \sum_{t=1}^T \widehat{r}_t$ $\rho = T \mu^* - \sum_{t=1}^T \widehat{r}_t$, where μ^* μ^* is the maximal reward mean, $\mu^* = \max_k \{\mu_k\}$ $\mu^* = \max_k \{\mu_k\}$, and $\widehat{r}_t$ $\widehat{r}_t$ is the reward in round t t . A zero-regret strategy is a strategy whose average regret per round ρ / T

ρ / T tends to zero with probability 1 when the number of played rounds tends to infinity. Intuitively, zero-regret strategies are guaranteed to converge to a (not necessarily unique) optimal strategy if enough rounds are played.

>> Section 7

Exp3 Source: EXP3 is a popular algorithm for adversarial multiarmed bandits, suggested and analyzed in this setting by Auer et al. [2002b]. Recently there was an increased interest in the performance of this algorithm in the stochastic setting, due to its new applications to stochastic multi-armed bandits with side information [Seldin et al., 2011] and to multi-armed bandits in the mixed stochastic-adversarial setting [Bubeck and Slivkins, 2012]. The paper presented an empirical evaluation and improved analysis of the performance of the EXP3 algorithm in the stochastic setting, as well as a modification of the EXP3 algorithm capable of achieving "logarithmic" regret in stochastic environment.

>> Section 8

Garivier and Moulines derive some of the first results with respect to bandit problems where the underlying model can change during play. A number of algorithms were presented to deal with this case, including Discounted UCB and Sliding-Window UCB. A similar approach based on Thompson Sampling algorithm is the f-Discounted-Sliding-Window Thompson Sampling (f-dsw TS) proposed by Cavenaghi et al. The f-dsw TS algorithm exploits a discount factor on the reward history and an arm-related sliding window to contrast concept drift in non-stationary environments. Another work by Burtini et al. introduces a weighted least squares Thompson sampling approach (WLS-TS), which proves beneficial in both the known and unknown non-stationary cases.

>> Section 9

Example: Iterated prisoner's dilemma An example often considered for adversarial bandits is the iterated prisoner's dilemma. In this example, each adversary has two arms to pull. They can either Deny or Confess. Standard stochastic bandit algorithms don't work very well with these iterations. For example, if the opponent cooperates in the first 100 rounds, defects for the next 200, then cooperates in the following 300, etc. then algorithms such as UCB won't be able to react very quickly to these changes. This is because after a certain point sub-optimal arms are rarely pulled to limit exploration and focus on exploitation. When the environment changes the algorithm is unable to adapt or may not even detect the change.

>> Section 10

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Dueling bandit The dueling bandit variant was introduced by Yue et al. (2012) to model the exploration-versus-exploitation tradeoff for relative feedback. In this variant the gambler is allowed to pull two levers at the same time, but they only get a binary feedback telling which lever provided the best reward. The difficulty of this problem stems from the fact that the gambler has no way of directly observing the reward of their actions. The earliest algorithms for this problem were InterleaveFiltering and Beat-The-Mean. The relative feedback of dueling bandits can also lead to voting paradoxes. A solution is to take the Condorcet winner as a reference. More recently, researchers have generalized algorithms from traditional MAB to dueling bandits: Relative Upper Confidence Bounds (RUCB), Relative EXponential weighing (REX3), Copeland Confidence Bounds (CCB), Relative Minimum Empirical Divergence (RMED), and Double Thompson Sampling (DTS).

>> Section 11

Epsilon-greedy strategy: The best lever is selected for a proportion $1 - \epsilon$ of the trials, and a lever is selected at random (with uniform probability) for a proportion ϵ . A typical parameter value might be $\epsilon = 0.1$, but this can vary widely depending on circumstances and predilections. **Epsilon-first strategy:** A pure exploration phase is followed by a pure exploitation phase. For N trials in total, the exploration phase occupies ϵN trials and the exploitation phase $(1 - \epsilon) N$ trials. During the exploration phase, a lever is randomly selected (with uniform probability); during the exploitation phase, the best lever is always selected. **Epsilon-decreasing strategy:** Similar to the epsilon-greedy strategy, except that the value of ϵ decreases as the experiment progresses, resulting in highly explorative behaviour at the start and highly exploitative behaviour at the finish. **Adaptive epsilon-greedy strategy based on value differences (VDBE):** Similar to the epsilon-decreasing strategy, except that epsilon is reduced on basis of the learning progress instead of manual tuning (Tokic, 2010). High fluctuations in the value estimates lead to a high epsilon (high exploration, low exploitation); low fluctuations to a low epsilon (low exploration, high exploitation). Further improvements can be achieved by a softmax-weighted action selection in case of exploratory actions (Tokic & Palm, 2011). **Adaptive epsilon-greedy strategy based on Bayesian ensembles (Epsilon-BMC):** An adaptive epsilon adaptation strategy for reinforcement learning similar to VBDE, with monotone convergence guarantees. In this framework, the epsilon parameter is viewed as the

expectation of a posterior distribution weighting a greedy agent (that fully trusts the learned reward) and uniform learning agent (that distrusts the learned reward). This posterior is approximated using a suitable Beta distribution under the assumption of normality of observed rewards. In order to address the possible risk of decreasing epsilon too quickly, uncertainty in the variance of the learned reward is also modeled and updated using a normal-gamma model. (Gimelfarb et al., 2019).